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Inventor(s)	PARK; Seong Su

URINE ANALYSIS DEVICE

Abstract

A urine analysis device according to an embodiment of the present disclosure includes: a mounting unit in which a test kit is mounted, the test kit including a container into which urine is collected and a curved test strip which is attached along an inner or outer circumferential surface of the container and includes a plurality of test papers provided to test the collected urine; a plurality of sensor units, each composed of a pair of a light emitting unit configured to emit light onto each of the plurality of test papers of the test kit mounted on the mounting unit and a light receiving unit configured to detect a color of each test paper based on the light reflected from each of the plurality of test papers; and an analysis unit that analyzes the urine based on the color information of the test papers detected by the light receiving units.

Inventors: **PARK; Seong Su (Anseong-si, KR)**

Applicant: **DNC BIOTECHNOLOGY INC.** (Daejeon, KR)

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a urine analysis device.

BACKGROUND

[0002] In general, a urine test is an essential examination that occupies about 30% of the total number of examinations performed in most hospitals or clinics. The urine test is a method to examine the components of urine to assess systemic conditions and detect lesions.

[0003] A conventional urine test is performed to examine urine by bringing a test stick with a paper test strip into contact with an examinee's urine and checking a color change of the paper test strip. That is, an examiner performs the urine test by directly applying the examinee's urine to the test stick, or by collecting urine in a container, such as a paper cup, and bringing the test stick into contact with the collected urine.

[0004] When a urine test is performed by using a urine analysis device including a urine test kit equipped with a flat test strip, the test is performed by emitting light over the entire area of the flat test strip and receiving the reflected light to determine the color. However, when the test is performed by using a urine test kit equipped with a curved test strip, it is difficult to determine the color of the entire area of the curved test strip by the conventional test method.

[0005] Also, if urine test kits are manufactured with varying degrees of curvature, such as differences in the radius of curvature or radius, urine analysis devices capable of accurate detection are required for each type.

DISCLOSURE OF THE INVENTION

Problems to be Solved by the Invention

[0006] The present disclosure is conceived to provide a device for analyzing urine by using a urine test kit equipped with a curved test strip.

[0007] However, the problems to be solved by the present disclosure are not limited to the above-described problems. There may be other problems to be solved by the present disclosure.

Means for Solving the Problems

[0008] According to an aspect of the present disclosure, a urine analysis device includes: a mounting unit in which a test kit is mounted, the test kit including a container into which urine is collected and a curved test strip which is attached along an inner or outer circumferential surface of the container and includes a plurality of test papers provided to test the collected urine; a plurality of sensor units, each composed of a pair of a light emitting unit configured to emit light onto each of the plurality of test papers of the test kit mounted on the mounting unit and a light receiving unit configured to detect a color of each test paper based on the light reflected from each of the plurality of test papers; and an analysis unit that analyzes the urine based on the color information of the test papers detected by the light receiving units.

[0009] According to another aspect of the present disclosure, a urine analysis device includes: a mounting unit in which a test kit is mounted, the test kit including a container into which urine is collected and a curved test strip which is attached along an inner or outer circumferential surface of the container and includes a plurality of test papers provided to test the collected urine; a substrate unit provided on an inner surface of the mounting unit and formed of a ductile material; a sensor unit provided on the substrate unit and composed of a light emitting unit configured to emit light

onto the plurality of test papers and a light receiving unit configured to detect a color of the plurality of test papers based on the light reflected from the plurality of test papers; and an analysis unit that analyzes the urine based on the color information of the test papers detected by the light receiving unit.
[0010] The above-described aspects are provided by way of illustration only and should not be construed as liming the present disclosure. Besides the above-described embodiments, there may be additional embodiments described in the accompanying drawings and the detailed description.
Effects of the Invention
[0011] According to any one of the above-described means for solving the problems of the present disclosure, it is possible to more accurately perform a urine test even when a urine test kit equipped with a curved test strip is used.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0012] FIG. **1** is a schematic diagram illustrating a urine analysis device according to an embodiment of the present disclosure.
[0013] FIG. **2** is a configuration view of the urine analysis device according to an embodiment of the present disclosure.
[0014] FIG. **3** shows a urine test kit mounted on the urine analysis device according to an embodiment of the present disclosure.
[0015] FIG. **4** is a planar figure of a substrate unit of the urine analysis device according to an embodiment of the present disclosure.
[0016] FIG. **5** illustrates the state in which the substrate unit shown in FIG. **4** is disposed to surround the urine test kit.
[0017] FIG. **6**A is a diagram for explaining a method of acquiring color information of each of a plurality of test papers by emitting light from the urine analysis device according to an embodiment of the present disclosure.
[0018] FIG. **6**B is a diagram for explaining a method of acquiring color information of each of a plurality of test papers by emitting light from the urine analysis device according to an embodiment of the present disclosure.
[0019] FIG. **7** is a flowchart of a urine analysis method using the urine test kit according to an embodiment of the present disclosure.
BEST MODE FOR CARRYING OUT THE INVENTION
[0020] Hereinafter, embodiments will be described in detail with reference to the accompanying drawings so that the present disclosure may be readily implemented by a person with ordinary skill in the art. However, it is to be noted that the present disclosure is not limited to the embodiments but can be embodied in various other ways. In the drawings, parts irrelevant to the description are omitted for the simplicity of explanation, and like reference numerals denote like parts throughout the whole document.
[0021] Throughout the whole document, the term “connected to” or “coupled to” that is used to designate a connection or coupling of one element to another element includes both a case that an element is “directly connected or coupled to” another element and a case that an element is “electronically connected or coupled to” another element via still another element. Further, it is to be understood that the term “comprises or includes” and/or “comprising or including” used in the document means that one or more other components, steps, operation and/or existence or addition of elements are not excluded in addition to the described components, steps, operation and/or elements unless context dictates otherwise and is not intended to preclude the possibility that one or more other features, numbers, steps, operations, components, parts, or combinations thereof may

exist or may be added.
[0022] Throughout the whole document, the term “unit” includes a unit implemented by hardware or software and a unit implemented by both of them. One unit may be implemented by two or more pieces of hardware, and two or more units may be implemented by one piece of hardware.
[0023] In the present specification, some of operations or functions described as being performed by a device may be performed by a server connected to the device. Likewise, some of operations or functions described as being performed by a server may be performed by a device connected to the server.
[0024] Hereinafter, embodiments of the present disclosure will be described in detail with reference to the accompanying drawings.
[0025] FIG. **1** is a schematic diagram illustrating a urine analysis device according to an embodiment of the present disclosure.
[0026] Referring to FIG. **1**, a urine analysis device **200** according to an embodiment of the present disclosure is configured to analyze urine collected into a cup-type urine test kit **100** from an examinee by detecting the color of a test paper **135** of a curved test strip **130** which reacts with the urine of the examinee.
[0027] In an embodiment of the present disclosure, the urine test kit **100** may include a container **110** and the curved test strip **130** attached along an inner or outer circumferential surface of the container **110**.
[0028] Specifically, as shown in FIG. **1**, the urine test kit **100** may include the container **110** into which urine is collected, and the curved test strip **130** which is attached along the inner or outer circumferential surface of the container **110** and includes a test paper **135** provided to test the urine collected in the container **110**.
[0029] For example, the curved test strip **130** may be attached along the inner or outer circumferential surface of the container **110** with an appropriate curvature, depending on the size of the container **110** and the length of the curved test strip **130**.
[0030] The curved test strip **130** may individually accommodate test papers **135**, which react with urine, in separate compartments and may include flow paths that allow urine to flow into the respective compartments.
[0031] In this case, a plurality of test papers **135** may be formed corresponding in number to various test items. For example, the test papers **135** may be 11, 7, or 4 in number, as shown in Table 1.
TABLE-US-00001 TABLE 1

No.	Test Item	Initial Color	11-Point Kit	7-Point Kit	4-Point Kit
1	Occult Blood	Color a	∘	∘	∘
2	Bilirubin	Color b	∘	—	—
3	Urobilinogen	Color c	∘	∘	—
4	Ketones	Color d	∘	∘	—
5	Protein	Color e	∘	—	∘
6	Nitrite	Color f	∘	∘	—
7	Glucose	Color g	∘	∘	∘
8	pH	Color h	∘	∘	—
9	Specific Gravity	Color i	∘	—	—
10	Leukocytes	Color j	∘	∘	∘
11	Vitamin C	Color k	∘	—	—

[0032] Referring to Table 1, the test items may include occult blood (OBD), bilirubin (BIL), urobilinogen (URO), ketones (KET), protein (PRO), nitrite (NIT), glucose (GLU), pH, specific gravity (SG), leukocytes (LEU), and vitamin C. Urine analysis can be performed by using the urine test kit **100** which includes 11, 7, or 4 test papers **135** depending on the purpose of the test.
[0033] The flow path may be divided into a lower flow path connected to a lower side of the test paper **135** and an upper flow path connected to an upper side of the test paper **135**. When an air pump (not shown) attached to the upper flow path is pressed, urine collected in the container **110** flows through the lower flow path and wets the test paper **135**.
[0034] The container **110** may be cylindrical or cup-shaped and formed of a transparent plastic material to facilitate urine collection from the examinee and to allow visual inspection of the collected urine.
[0035] In an embodiment, when the container **110** is mounted in a mounting unit of the urine analysis device **200**, which will be described later, a protrusion or groove for guiding the container

110 to a correct mounting position may be formed at the bottom of the container **110**. Conversely, a groove or protrusion corresponding to the protrusion or groove of the container **110** may be formed in the mounting unit of the urine analysis device **200**.

[0036] Meanwhile, a tag or code for identifying the type of the urine test kit **100** and recording information of the examinee may be attached to one surface of the container **110**.

[0037] For example, an NFC reader (not shown) may be attached to the one surface of the container **110**. The NFC reader is configured to read recorded data from an NFC tag attached to the urine test kit **100**. The NFC tag may record information, such as the examinee's data and the type of urine test kit **100** indicating whether it is a 11-point kit, a 7-point kit, or a 4-point kit. Although the NFC tag has been described as an example in the embodiment of the present disclosure, RFID, barcodes, or QR codes may also be used.

[0038] The urine analysis device **200** according to an embodiment of the present disclosure may be a device that analyzes urine by using the urine test kit **100** equipped with the curved test strip **130**.

[0039] The urine analysis device **200** of the present embodiment may be configured to allow the urine test kit **100** to be mounted. Once the urine test kit **100** is mounted, the urine analysis device **200** emits light onto the test papers **135** of the curved test strip **130**, receives light reflected from the test papers **135**, and analyzes the urine based on color information of the reflected light. The urine analysis device **200** may include a display to present the analysis result of the urine.

[0040] The urine analysis device **200** according to the present embodiment can accurately perform a urine test even when the urine test kit **100** equipped with the curved test strip **130** is used.

[0041] Further, the urine analysis device **200** according to an embodiment of the present disclosure can adjust individual test times and produce an accurate test result through scanning tailored to each test item.

[0042] For example, 30 seconds may be required for glucose and bilirubin, 40 seconds may be required for ketones, 45 seconds may be required for specific gravity, 60 seconds may be required for pH, protein, urobilinogen, blood and nitrite, and 2 minutes may be required for leukocytes. In this case, pH and protein can be read immediately or after more than 2 minutes, and the result for leukocytes can be confirmed based on a reaction that occurs after a delay of about 1 to 2 minutes.

[0043] Hereafter, the urine analysis device **200** according to an embodiment of the present disclosure will be described in detail with reference to FIG. **2** to FIG. **6**B.

[0044] FIG. **2** is a configuration view of the urine analysis device according to an embodiment of the present disclosure, and FIG. **3** shows the urine test kit **100** mounted on the urine analysis device **200** according to an embodiment of the present disclosure.

[0045] Referring to FIG. **2**, the urine analysis device **200** according to the present embodiment may include a mounting unit **210**, sensor units **220** and **230**, an analysis unit **240**, a measurement unit **250**, and a correction unit **260**. However, the urine analysis device **200** illustrated in FIG. **2** is merely an embodiment of the present disclosure, and various modifications based on the components illustrated in FIG. **2** are possible. Herein, the measurement unit **250** and the correction unit **260** may be implemented by an MCU with built-in memory and peripherals. The built-in memory may store firmware and a standard colorimetric table containing reference color values for each test item.

[0046] As shown in FIG. **3**, the urine test kit **100** may be mounted in the mounting unit **210**. That is, the mounting unit **210** may include a predetermined space in which the urine test kit **100** can be mounted.

[0047] In this case, the mounting unit **210** may be formed to a height sufficient to cover the test paper **135** of the urine test kit **100**. For example, the mounting unit **210** may have a substantially cylindrical space capable of accommodating the entire urine test kit **100**. For example, the accommodating space may have a cylindrical shape concentric with the urine test kit **100** and thus can facilitate alignment between the sensor unit and the curved test strip **130**.

[0048] After the urine test kit **100** is accommodated in the accommodating space, the

accommodating space can be sealed with an upper cover (see FIG. **3**) to suppress the introduction of external light into the accommodating space.
[0049] Furthermore, although not illustrated in the drawings, a groove or protrusion corresponding to the protrusion or groove of the container **110** may be formed in the mounting unit **210**, and, thus, the urine test kit **100** can be accurately positioned in the mounting unit **210** to allow precise sensing by the sensor unit to be described below.
[0050] In this case, the sensor unit and the curved test strip **130** may be arranged as concentric circles with different radii, assuming that each of the sensor unit and the curved test strip **130** is extended with the same curvature. Accordingly, each test paper **135** of the curved test strip **130** can be aligned to face a corresponding sensor unit.
[0051] A plurality of sensor units may be provided on an inner surface of the mounting unit **210**, and each sensor unit may include a pair of a light emitting unit **220** and a light receiving unit **230**.
[0052] For example, the sensor units may be arranged on at least a part of an inner wall of the mounting unit **210** and oriented toward the center of the accommodating space. Thus, when the urine test kit **100** is mounted in the mounting unit **210**, the sensor units are aligned to face the curved test strip **130** and can scan the test papers **135**.
[0053] In an embodiment, the sensor units are formed on a flexible printed circuit board (FPCB) and can be easily positioned on the inner wall of the mounting unit **210**. That is, the FPCB may be disposed on at least a part of an inner wall in the cylindrical accommodating space of the mounting unit **210**, and the sensor units may be formed on the FPCB disposed on the inner wall in the cylindrical accommodating space.
[0054] The light emitting unit **220** may emit light onto the test paper **135** of the test strip **130** provided in the urine test kit **100**, and may be configured as, for example, a white light LED sensor. For example, the light emitting unit **220** may be composed of a high-brightness white LED driven by a constant current control method and a MOSFET capable of changing a current applied to the LED in response to a PWM control signal from the MCU.
[0055] In an embodiment, the LED may be a high-brightness white LED with a color temperature of 5600 K.
[0056] As described above, the sensor units are aligned to face the curved test strip **130**, and, thus, a plurality of light emitting units **220** is positioned to correspond to the test papers **135** included in the curved test strip **130** and emits light onto the test papers **135**. Thus, it is possible to provide sufficient illuminance to the test papers **135**.
[0057] Further, the light receiving unit **230** may receive light reflected from the test papers **135**. For example, the light receiving unit **230** may be configured as a light receiving sensor with a 16-bit resolution.
[0058] The light receiving unit **230** may be an image sensor configured to detect the color of the test paper **135** of the curved test strip **130** based on the light reflected from the test paper **135**. For example, the light receiving unit **230** may be an RGB color sensor capable of detecting red, green and blue light, or an RGBW color sensor, such as VEML6040, capable of detecting red, green, blue and white light, using an I.sup.2C interface.
[0059] The VEML6040 may be a component having L×W×H of 2.0 mm×1.25 mm×1.0 mm, with each channel (R, G, B, W) having a 16-bit resolution.
[0060] Furthermore, the light receiving unit **230** and the MCU may be connected by an I.sup.2C bus, which allows the mater MCU to read RGBW data detected by the slave light receiving unit **230**.
[0061] The light emitting unit **220** and the light receiving unit **230** may be appropriately arranged or separated to suppress the direct introduction of light into the light receiving unit **230**. Therefore, each light receiving unit **230** may be configured to receive only the reflected light of the light emitted from the corresponding light emitting unit **220**.
[0062] Meanwhile, the urine analysis device **200** of the present embodiment may further include a

substrate unit **201** that supports the sensor units **220** and **230**. That is, the substrate unit **201** may be provided on the inner surface of the mounting unit **210**, and a plurality of sensor units **220** and **230** may be arranged on the substrate unit **201** at predetermined intervals.

[0063] In the present embodiment, the substrate unit **201** may be configured as an FPCB to be arranged on the inner wall of the mounting unit **210** with the same or similar curvature as the test strip **130**, which is bent along the circumference of the container **110**. The substrate unit **201** may be configured to read various types of samples and formed with sufficient flexibility and thinness to maintain the pliability of the material.

[0064] After the urine test kit **100** is mounted in the mounting unit **210**, the analysis unit **240** according to an embodiment of the present disclosure may identify the type of urine test kit **100** (11-point kit, 7-point kit, or 4-point kit) based on the number of test items.

[0065] The correction unit **260** may select available sensor units from among the plurality of sensor units **220** and **230** based on data received from the plurality of sensor units **220** and **230**.

[0066] In an embodiment, data related to optical signals received by the plurality of sensor units (e.g., the light receiving units **230**) may be transmitted to the correction unit **260**. The correction unit **260** may select available sensor units **220** and **230** positioned to best detect the test paper **135** of the urine test kit **100** from among the plurality of densely arranged sensor units **220** and **230**.

[0067] Table 2 below shows matching data including available sensor units with color sensors for each test paper **135** when the urine test kit **100** (11-point kit) includes 11 test papers **135** in the test strip **130**.

TABLE-US-00002 TABLE 2 No. Test Item Test Paper Available Sensor Unit 1 UT1 TP1 Light Emitting Unit 1, Light Receiving Unit 1 2 UT2 TP2 Light Emitting Unit 2, Light Receiving Unit 2 3 UT3 TP3 Light Emitting Unit 3, Light Receiving Unit 3 4 UT4 TP4 Light Emitting Unit 4, Light Receiving Unit 4 5 UT5 TP5 Light Emitting Unit 5, Light Receiving Unit 5 6 UT6 TP6 Light Emitting Unit 6, Light Receiving Unit 6 7 UT7 TP7 Light Emitting Unit 7, Light Receiving Unit 7 8 UT8 TP8 Light Emitting Unit 8, Light Receiving Unit 8 9 UT9 TP9 Light Emitting Unit 9, Light Receiving Unit 9 10 UT10 TP10 Light Emitting Unit 10, Light Receiving Unit 10 11 UT11 TP11 Light Emitting Unit 11, Light Receiving Unit 11

[0068] In the present embodiment, the analysis unit **240** analyzes urine based on data received from the available sensor units. Specifically, the urine can be analyzed based on color information received from the sensor units **220** and **230**, and as described above, the test time can be set according to each test item.

[0069] When the urine test kit **100** is mounted in the mounting unit **210**, the analysis unit **240** sequentially detects the colors of the test papers sensed by the available sensor units based on the data of the available sensor units. Then, the analysis unit **240** compares the detected colors with reference colors for the corresponding test items in the standard colorimetric table, determines the levels of the test items, and derives the test result by analyzing the test items and information of the examinee.

[0070] Hereafter, the test strip **130** including, for example, less than 11 test papers **135** will be described.

[0071] FIG. **4** is a planar figure of the substrate unit **201** of the urine analysis device **200** according to an embodiment of the present disclosure.

[0072] In the urine analysis device **200** according to an embodiment of the present disclosure, the number of sensor units **220** and **230** may be set greater than the number of test papers **135** of the test strip **130** in the urine test kit **100**.

[0073] Specifically, as shown in FIG. **4**, the sensor units **220** and **230** may be arranged at predetermined intervals along one direction of the substrate unit **201**, for example, a circumferential direction of the mounting unit **210** or the container **110**. For example, the number of test papers **135** in the test strip **130** used in the urine analysis device **200** of the present embodiment may be 11, 7, or 4, and the number of sensor units **220** and **230** may be 22 (11 light

emitting units and 11 light receiving units).
[0074] Also, as shown in FIG. **4**, the plurality of sensor units **220** and **230** may be densely arranged such that a gap G between the sensor units **220** and **230** is smaller than a width W of each individual sensor unit in one direction. Further, the density of the sensor units **220** and **230** arranged in the space may be set higher than that of the test papers **135**.
[0075] Since the sensor units **220** and **230**, which are greater in number than the test papers **135**, are densely arranged as described above, it is possible to analyze various urine test kits **100** with different positions or numbers of test papers **135**. Therefore, without the need for separate devices for different urine test kits **100** varying in the types or numbers of test items and the sizes of containers **110**, the urine analysis device **200** according to the present embodiment can effectively perform analysis.
[0076] FIG. **5** illustrates the state in which the substrate unit **201** shown in FIG. **4** is disposed to surround the urine test kit **100**.
[0077] In an embodiment of the present disclosure, the substrate unit **201** may be formed to allow deformation. The substrate unit **201**, which may be formed of a ductile material, may include notches between the sensor units **220** and **230** to allow stretching or deformation in one direction. As the substrate unit **201** is stretched or deformed, a gap between the plurality of sensor units **220** and **230** may vary.
[0078] To enable stretching of the substrate unit **201**, the urine analysis device **200** of the present embodiment may further include a connection unit **202**. As shown in FIG. **5**, both ends of the connection unit **202** may be fixed respectively to the mounting unit **210** and the substrate unit **201**. Also, the connection unit **202** may be elastically deformed in length by user manipulation or applied force and thus can pull or push the substrate unit **201** in one direction.
[0079] For example, when the substrate unit **201** is pulled, it may elongate within an allowable range, which increases the gap between the plurality of sensor units **220** and **230**. When the pulling force is released or a pushing force is applied toward the substrate unit **201**, the gap between the plurality of sensor units **220** and **230** may be decreased. Further, by pulling or pushing the substrate unit **201**, the substrate can be brought into close contact with the container **110** so that the distance between the sensor units **220** and **230** and the test papers **135** is suitable for analysis.
[0080] In the present embodiment, by arranging the plurality of sensor units **220** and **230** at a high density and varying the gap between the sensor units **220** and **230** in the urine analysis device **200**, it is possible to analyze various urine test kits **100** with different sizes or curvature radii of the container **110** and different numbers or positions of test papers **135**. Furthermore, the analysis can still be performed even if there is a variation in the mounting position of the urine test kit **100** in the mounting unit **210**.
[0081] FIG. **6**A and FIG. **6**B are diagrams for explaining methods of acquiring color information of each of the plurality of test papers **135** by emitting light from the urine analysis device **200** according to an embodiment of the present disclosure.
[0082] In FIG. **6**A and FIG. **6**B, the sensor unit is illustrated as having a configuration in the light emitting unit **220** and the light receiving unit **230** are integrally formed, but is not limited thereto. The light emitting sensor and the light receiving sensor may also be separately formed.
[0083] As shown in the drawings, some of the plurality of sensor units **220** and **230**, which is densely arranged relative to the test papers **135**, may be positioned to correspond to at least one of the test papers **135** included in the curved test strip **130**.
[0084] As shown in FIG. **6**A, the light emitting unit **220** may emit light toward the corresponding test strip **130**. Further, as shown in FIG. **6**B, the light receiving unit **230** may receive light reflected from the corresponding test paper **135**.
[0085] In the present embodiment, even if all the light emitting units **220** emit light toward the test strip **130**, the light receiving unit located at a position corresponding to the test paper **135** receives light effective for analysis. Thus, the sensor unit with the light receiving unit positioned to

correspond to the test paper **135** can serve as an available sensor unit. FIG. **6**A and FIG. **6**B illustrate light emission and reception in the available sensor unit. However, before the available sensor unit is selected, all the light emitting units **220** and light receiving units **230** may operate simultaneously.

[0086] Table 3 below shows matching data including available sensor units with color sensors for each test paper **135** when the urine test kit **100** (7-point kit) includes 7 test papers **135** in the test strip **130**.

TABLE-US-00003 TABLE 3 Available/ Color Sensor Unit Unavailable Value Test Pad Test Item Light Emitting Unit 1, Available a TP1 UT1 Light Receiving Unit 1 Light Emitting Unit 2, Unavailable — — — Light Receiving Unit 2 Light Emitting Unit 3, Available b TP2 UT2 Light Receiving Unit 3 Light Emitting Unit 4, Unavailable — — — Light Receiving Unit 4 Light Emitting Unit 5, Available c TP3 UT3 Light Receiving Unit 5 Light Emitting Unit 6, Available d TP4 UT4 Light Receiving Unit 6 Light Emitting Unit 7, Available e TP5 UT5 Light Receiving Unit 7 Light Emitting Unit 8, Unavailable — — — Light Receiving Unit 8 Light Emitting Unit 9, Available f TP6 UT6 Light Receiving Unit 9 Light Emitting Unit 10, Unavailable — — — Light Receiving Unit 10 Light Emitting Unit 11, Available g TP7 UT7 Light Receiving Unit 11

[0087] Table 4 below shows matching data including available sensor units with color sensors for each test paper **135** when the urine test kit **100** (4-point kit) includes 4 test papers **135** in the test strip **130**.

TABLE-US-00004 TABLE 4 Available/ Color Sensor Unit Unavailable Value Test Pad Test Item Light Emitting Unit 1, Unavailable — — — Light Receiving Unit 1 Light Emitting Unit 2, Available a TP1 UT1 Light Receiving Unit 2 Light Emitting Unit 3, Unavailable — — — Light Receiving Unit 3 Light Emitting Unit 4, Unavailable — — — Light Receiving Unit 4 Light Emitting Unit 5, Available b TP2 UT2 Light Receiving Unit 5 Light Emitting Unit 6, Unavailable — — — Light Receiving Unit 6 Light Emitting Unit 7, Unavailable — — — Light Receiving Unit 7 Light Emitting Unit 8, Available c TP3 UT3 Light Receiving Unit 8 Light Emitting Unit 9, Unavailable — — — Light Receiving Unit 9 Light Emitting Unit 10, Unavailable — — — Light Receiving Unit 10 Light Emitting Unit 11, Available d TP4 UT4 Light Receiving Unit 11

[0088] According to an embodiment of the present disclosure, even if the urine test kit **100** varies in type (11-point kit, 7-point kit, or 4-point kit), urine analysis can be performed just by mounting the urine test kit **100** in the urine analysis device **200** without the need for a separate matching process between the color sensors and the test papers.

[0089] Therefore, the urine analysis device **200** of the present disclosure provides excellent compatibility since it is possible to relatively freely design of the number and size of test papers **135** in the urine test kit **100**.

[0090] Meanwhile, in the urine analysis device **200** according to an embodiment of the present disclosure, a color value of a sample to be analyzed may be affected and changed due to changes in temperature or humidity of the sensor units **220** and **230** and its surroundings. The urine analysis device **200** according to an embodiment of the present disclosure can enhance the accuracy of analysis by measuring temperature or humidity through the measurement unit **250** and either maintaining the value within a predetermined range or compensating for any variations.

[0091] Specifically, the measurement unit **250** may measure the temperature of the mounting unit **210**, the sensor units **220** and **230**, the substrate unit **201**, or its surroundings via a temperature sensor (not shown). Further, the correction unit **260** may correct a duty ratio of light emitted by the sensor units **220** and **230** based on the temperature measured by the measurement unit **250**.

[0092] That is, when the light emitting unit **220** continuously emits light and increases in temperature, the correction unit **260** may increase a duty ratio of the light emitting unit **220** via PWM control. For example, by increasing the duty ratio by 2% for every 2° C. increase in temperature, the output of the light emitting unit **220** can be increased at higher temperatures and deviations in optical data caused by temperature changes can be compensated.

[0093] The measurement unit **250** may detect whether external light enters the accommodating space of the mounting unit **210** while the urine test kit **100** is mounted, using an illuminance sensor (not shown) provided within the accommodating space. When the external light is detected entering the accommodating space, an alarm can be generated via a speaker (not shown), display, or the like.
[0094] Also, the measurement unit **250** may measure the humidity of the mounting unit **210**, the sensor units **220** and **230**, the substrate **201**, or its surroundings. The correction unit **260** may compare the measured humidity measured by the measurement unit **250** with a predetermined value or range. When the humidity is not equal to a desirable value or within a desirable range, the urine analysis device **200** can generate an alarm, which prompts the user to take corrective action, such as ventilation, to bring the humidity within the desirable range.
[0095] Further, the correction unit **260** of the urine analysis device **200** according to the present embodiment may calculate the distance between the sensor units **220** and **230** and the test papers **135**. In the urine analysis device **200** of the present embodiment, the container **110** may be inaccurately placed in the mounting unit **210** depending on user manipulation. In response, the correction unit **260** may calculate the distance between the sensor units **220** and **230** and the test papers **135** based on the data received from the sensor units **220** and **230**. The correction unit **260** may compare the calculated distance with a predetermined value or range. When the distance is not equal to a desirable value or within a desirable range, the urine analysis device **200** of the present embodiment can generate an alarm, which prompts the user to take corrective action. For example, the user may be guided to place the urine test kit **100** in the mounting unit **210** again.
[0096] Meanwhile, the urine analysis device **200** according to an embodiment of the present disclosure may further include an output unit. For example, the urine analysis device **200** may be equipped with a display to present to present the analysis result of the tested urine. The output unit can also output the above-described alarm to the user via visual or auditory information. Further, the output unit can transmit the urine analysis result and other data to a device through a communication means.
[0097] FIG. **7** is a flowchart of a urine analysis method using the urine test kit **100** according to an embodiment of the present disclosure. The urine analysis method performed by the urine analysis device **200** illustrated in FIG. **7** includes the processes time-sequentially performed by the urine analysis device **200** illustrated in FIG. **1** to FIG. **6**B. Therefore, the descriptions of the processes may also be applied to the urine analysis method performed by the urine analysis device **200** according to the embodiment illustrated in FIG. **1** to FIG. **6**B even though they are omitted hereinafter.
[0098] In a process S**710**, a urine test kit may be prepared. The urine test kit **100** in which a patient's urine has been collected may be positioned in the mounting unit **210** of the urine analysis device **200**.
[0099] In a process S**720**, a plurality of sensor units may emit light onto test papers of a test strip included in the urine test kit and receive light reflected from the test papers. The plurality of sensor units **220** and **230** may emit and receive light either simultaneously or sequentially.
[0100] In a process S**730**, available sensor units are selected from among the plurality of sensor units. Based on the data received from the plurality of sensor units **220** and **230**, sensor units that receive valid data suitable for urine analysis may be selected as available sensor units.
[0101] In a process S**740**, urine is analyzed based on data received from the available sensor units. The components of urine may be analyzed based on color information in the received data.
[0102] In a process S**750**, the analysis result of the urine may be output. The analysis result may be output to the user via an output unit such as a display.
[0103] Meanwhile, in an embodiment of the present disclosure, the urine test kit **100** may include a reference kit to select available sensor units. The reference kit may be a urine test kit which has a specific number and arrangement of test papers and in which urine has not been collected.
[0104] In the process S**730** of selecting available sensor units, the data received from the plurality

of sensor units **220** and **230** regarding the test papers **135** of the test strip **130** included in the reference kit may be used as a criterion for selecting available sensor units. The reference kit may be used to select available sensor units by comparing the color information received from the sensor units or comparing color information depending on the positions of a substrate unit and test papers.

[0105] As described above, the urine analysis device according to an embodiment of the present disclosure includes sensor units, which are greater in number than the test papers in the urine test kit, and, thus. Therefore, sensor units at certain positions in one direction (for example, odd-numbered sensor units, see FIG. **6**A and FIG. **6**B) can be selected as available sensor units. Alternatively, sensor units up to a certain position (for example, up to the 11th sensor unit) may be selected as available sensor units.

[0106] The urine analysis method of the present embodiment may further include a process of correcting a duty ratio of light emitted by the sensor unit based on the temperature of the sensor unit or its surroundings (S**715**). In this case, the temperature may be a value measured by the above-described measurement unit **250**.

[0107] The urine analysis method of the present embodiment may also include a process of generating an alarm signal when the measured humidity of the sensor unit or its surroundings does not satisfy a predetermined desirable value or range by comparing it with the predetermined value or range (S**715**). The humidity may be measured by the above-described measurement unit **250**, and the alarm output by the output unit may prompt the user to take appropriate action.

[0108] The urine analysis method of the present embodiment may further include a process of calculating the distance between the sensor unit and the test paper based on the data received from the sensor unit (S**725**). The calculated distance may be compared with a predetermined value or range, and an alarm signal may be generated according to predetermined conditions to prompt the user to take appropriate action. By prompting the user to perform manipulation, it is possible to suppress the reception of light or color information that is unsuitable for urine analysis.

[0109] In the descriptions above, the processes S**710** to S**750** may be divided into additional processes or combined into fewer processes depending on an embodiment. In addition, some of the processes may be omitted and the sequence of the processes may be changed if necessary.

[0110] The urine analysis method illustrated in FIG. **7** can be implemented as a computer program stored in a medium to be executed by a computer or a storage medium including instructions executable by a computer. Also, the urine analysis method illustrated in FIG. **7** can be implemented as a computer program stored in a medium to be executed by a computer.

[0111] A computer-readable medium can be any usable medium which can be accessed by the computer and includes all volatile/non-volatile and removable/non-removable media. Further, the computer-readable medium may include all computer storage media. The computer storage media include all volatile/non-volatile and removable/non-removable media embodied by a certain method or technology for storing information such as computer-readable instruction code, a data structure, a program module or other data.

[0112] The above description of the present disclosure is provided for the purpose of illustration, and it would be understood by a person with ordinary skill in the art that various changes and modifications may be made without changing technical conception and essential features of the present disclosure. Thus, it is clear that the above-described examples are illustrative in all aspects and do not limit the present disclosure. For example, each component described to be of a single type can be implemented in a distributed manner. Likewise, components described to be distributed can be implemented in a combined manner.

[0113] The scope of the present disclosure is defined by the following claims rather than by the detailed description of the embodiment. It shall be understood that all modifications and embodiments conceived from the meaning and scope of the claims and their equivalents are included in the scope of the present disclosure.

Claims

1. A urine analysis device, comprising: a mounting unit in which a test kit is mounted, the test kit including a container into which urine is collected and a curved test strip which is attached along an inner or outer circumferential surface of the container and includes a plurality of test papers provided to test the collected urine; a plurality of sensor units, each composed of a pair of a light emitting unit configured to emit light onto each of the plurality of test papers of the test kit mounted on the mounting unit and a light receiving unit configured to detect a color of each test paper based on the light reflected from each of the plurality of test papers; and an analysis unit that analyzes the urine based on the color information of the test papers detected by the light receiving units.
2. The urine analysis device of claim 1, wherein each of the sensor units performs scanning tailored to test items associated with its corresponding test paper.
3. The urine analysis device of claim 1, further comprising: a correction unit configured to measure a temperature of the mounting unit or the plurality of sensor units via a temperature sensor and correct a duty ratio of light emitted by the light emitting unit based on the measured temperature.
4. The urine analysis device of claim 1, wherein a protrusion or groove configured to guide the container to a mounting position is formed at the bottom of the container when the mounting unit is mounted, and a groove or protrusion corresponding to the protrusion or groove of the container is formed in the mounting unit.
5. The urine analysis device of claim 1, wherein the number of the plurality of sensor units is different from the number of the plurality of test papers.
6. The urine analysis device of claim 1, wherein the number of the plurality of sensor units is equal to the number of the plurality of test papers.
7. The urine analysis device of claim 1, wherein the mounting unit is formed to a height sufficient to cover the plurality of test papers.
8. The urine analysis device of claim 1, wherein the mounting unit is formed to a height sufficient to accommodate the entire urine test kit.
9. The urine analysis device of claim 8, wherein after the urine test kit is mounted in the mounting unit, an accommodating space of the mounting unit is sealed with an upper cover.
10. The urine analysis device of claim 1, wherein an accommodating space of the mounting unit has a cylindrical shape concentric with the urine test kit.
11. The urine analysis device of claim 10, wherein the plurality of sensor units is formed on a substrate provided on at least a part of an inner wall in the cylindrical accommodating space and formed of a ductile material.
12. The urine analysis device of claim 1, wherein the plurality of sensor units is formed on a substrate provided on an inner wall of the mounting unit with the same curvature as the curved test strip, which is bent along a circumference of the container, and formed of a ductile material.
13. The urine analysis device of claim 1, wherein the plurality of sensor units is aligned to face the test papers of the curved test strip.
14. The urine analysis device of claim 1, wherein the plurality of sensor units is arranged on at least a part of an inner wall of the mounting unit and oriented toward a center of an accommodating space of the mounting unit.
15. A urine analysis device, comprising: a mounting unit in which a test kit is mounted, the test kit including a container into which urine is collected and a curved test strip which is attached along an inner or outer circumferential surface of the container and includes a plurality of test papers provided to test the collected urine; a substrate unit provided on an inner surface of the mounting unit and formed of a ductile material; a sensor unit provided on the substrate unit and composed of a light emitting unit configured to emit light to the plurality of test papers and a light receiving unit

configured to detect a color of the plurality of test papers based on the light reflected from the plurality of test papers; and an analysis unit that analyzes the urine based on the color information of the test papers detected by the light receiving unit.

16. The urine analysis device of claim 15, wherein the substrate unit formed of the ductile material is provide on an inner wall of the mounting unit with the same curvature as the curved test strip, which is bent along a circumference of the container.

17. The urine analysis device of claim 15, wherein an accommodating space of the mounting unit has a cylindrical shape concentric with the urine test kit.

18. The urine analysis device of claim 17, wherein the substrate unit formed of the ductile material is provided on at least a part of an inner wall in the cylindrical accommodating space.